

AI Research Workflow: Perplexity, Grok/X, Social Sentiment, and Verification

A source-discipline module for using AI as a research assistant, not a trading signal machine.

Module purpose

Turn the AI-tools discussion into a student workflow for market briefings, company research, social sentiment, verification, and fraud awareness.

Ideal length	Audience	Student output
90 minutes	Research-focused trading students	AI research log + verification checklist

Conversation source

Prepared from the April 21, 2026 class transcript. External references were added to convert the live discussion into a student-safe tutorial.

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1. Lesson snapshot

The conversation compared Perplexity, Comet, Grok/X, ChatGPT, Gemini, Claude, screeners, social sentiment, AI workers, and routines. The student-ready lesson is simple: AI can accelerate research, but it must not replace verification, risk management, or primary-source thinking.

Conversation idea	Student-ready version	Guardrail
Ask Perplexity for business news.	Use AI to create a first-pass briefing, then verify important claims with primary sources or reputable news.	No uncited AI answer goes into the trade plan.
Use Grok/X for social sentiment.	Social posts can show attention, narrative, and crowd emotion; they do not prove financial truth.	Sentiment is an input, not a signal.
AI workers scan the day.	Automations can remind, summarize, and organize. Students still decide whether a source is reliable.	Never delegate account access or trading authority without strict controls.
Different students need different tools.	Match the tool to the student's current skill: news summary first, then filings, then sentiment, then automation.	Do not overwhelm beginners with too many dashboards.

Learning objectives

- Build a repeatable AI research workflow for market preparation.
- Separate news summary, primary-source research, sentiment, and verification.
- Use social sentiment without treating it as a buy/sell signal.
- Create prompts that force bullish, bearish, and source-checking views.
- Recognize AI fraud, fake AI claims, deepfakes, and promotional manipulation.
- Maintain a research log with citations and uncertainty notes.

2. Tool map: what each tool is for

Tool type	Best use	Not good for
Perplexity / Comet	Morning brief, web research, broad current-event summaries, browser-assisted organization.	Final authority on filings, medical facts, or trade entries without verification.
Grok / X Search	Real-time social narrative, posts, threads, rumor detection, crowd focus.	Truth verification by itself.
ChatGPT Search / Deep Research	Structured reports, synthesis, source-backed summaries, uploaded-file analysis.	Live trading commands or uncited certainty.
EDGAR and company IR	Primary company filings, earnings releases, risk factors, investor presentations.	Social mood or minute-by-minute narrative.
Regulators: SEC, FINRA, CFTC, FDA	Rules, warnings, fraud alerts, product/regulatory risk.	Short-term price prediction.

Verification ladder
Primary source first. Regulator second. Reputable data/news third. Social media last. AI is the organizer, not the source of truth.

3. A source-backed AI workflow

Step	Prompt/task	Required output
1. Define question	What exactly am I trying to know before market open?	One sentence research question.
2. Choose source type	Is this a filing, news, price-data, social-sentiment, or regulatory question?	Correct source category.
3. Ask AI	Summarize the latest evidence and include sources.	Bullets with citations or source links.
4. Challenge AI	What could be wrong, stale, promotional, or missing?	Bear case and uncertainty list.
5. Verify	Open the source and check the claim.	Research log entry with link and date.
6. Decide use	Does this change my watchlist, chart level, or risk plan?	Action category: watch, ignore, research more, or avoid.

4. Prompt templates

Use case	Prompt
Morning brief	Give me the most important market catalysts for today. Separate confirmed events, scheduled events, and rumors. Cite sources.
Company thesis	Create bullish and bearish arguments for [ticker]. Use only recent filings, investor relations materials, and regulator sources first.
Social sentiment	Analyze current X discussion for [ticker/topic]. Separate volume of attention, repeated narratives, credible sources, and suspicious promotion.
Verification	List the five claims in this AI answer most likely to be wrong or outdated. Tell me what primary source would verify each one.
Student translation	Explain this term in simple English, then give a one-sentence Russian translation, then give a trading example.

Fraud and hype warning

AI can create fake confidence. Regulators warn that fraudsters use AI themes, deepfake audio/video, and fake claims about AI capabilities to lure investors.

5. Social sentiment: useful but dangerous

Social sentiment can reveal what people are paying attention to, what words are spreading, and whether a narrative is early or crowded. It cannot prove earnings quality, regulatory approval, accounting truth, or future price. The teacher should frame social sentiment as a heat map of attention, not a signal machine.

Signal to observe	What it may mean	What must be checked
Post volume spike	Attention is rising.	Is there real news, a filing, or only promotion?
Repeated phrase or meme	Narrative is spreading.	Who started it and is there evidence?
Influencer concentration	Few accounts may be driving the move.	Are accounts disclosed, conflicted, bot-like, or coordinated?
Sentiment flip	Crowd mood is changing.	Did the company facts change or only price action?
Screenshots/claims	Could be useful leads.	Verify against original documents and timestamps.

6. Student worksheet: AI research log

Pick one ticker or market theme. Complete this log before discussing any trade idea. The answer must include source links and a written uncertainty note.

Research question	AI tool used	Best source found	Bull case	Bear case	What might be false?	Decision category

Completion standard

A passing student can show where each important claim came from and can explain what evidence would change their mind.

External source stack for students

The source titles below are clickable in most PDF readers. They were selected because they are official, regulator-backed, exchange-backed, or institutionally useful for student verification.

ID	Source	Why it belongs in the tutorial
S1	Perplexity - Comet Browser perplexity.ai	Official page describing Comet as an AI browser/personal assistant for web research and task support.
S2	xAI Docs - X Search docs.x.ai	Explains Grok's X Search tool for keyword search, semantic search, user search, and thread fetch on X.
S3	X Developer Docs - Recent Search docs.x.com	Official API documentation for retrieving recent posts matching a query.
S4	OpenAI Help - ChatGPT Search help.openai.com	Explains source/citation behavior for ChatGPT search results.
S5	OpenAI Help - Deep Research in ChatGPT help.openai.com	Describes planning, researching, synthesizing, and producing sourced reports.
S6	Investor.gov - AI and Investment Fraud investor.gov	SEC/NASAA/FINRA investor alert warning about AI-enabled investment scams and deepfakes.
S7	FINRA - AI and Investment Fraud finra.org	Investor protection guidance on AI-themed fraud.
S8	FINRA - Social Media-Influenced Investing finra.org	Research background on social media tools, sentiment analysis, and influence on investment decisions.